of ordered pairs $\{(a,1),(b,2),\ldots,(z,26)\}$. This set of ordered pairs is a relation. The domain of the relation is $\{a,b,\ldots z\}$ and the range is $\{1,2,\ldots,26\}$.

Now we recall that the relations discussed in Illustrations 1.1 and 1.2 also end up with subsets of the cartesian product of two sets. So the term *relation* used in all discussions we had so far, fits with the mathematical term *relation* defined in Definition 1.2.

The domain of the relation discussed in Illustration 1.1 is the set $\{L,E,T,U,S,W,I,N\}$ and the range is $\{O,H,W,X,V,Z,L,Q\}$. In Illustration 1.2, the domain and range of the relation discussed for the equation $2x-y=0$ are $\mathbb{R}$ and $\mathbb{R}$ (See Figure 1.9); for the equation $x^2-y=0$, the domain is $\mathbb{R}$ and the range is $[0,\infty)$ (See Figure 1.10); and in the case of the third equation $x-y^2=0$, the domain is $[0,\infty)$ and the range is $\mathbb{R}$ (See Figures 1.11 and 1.12).

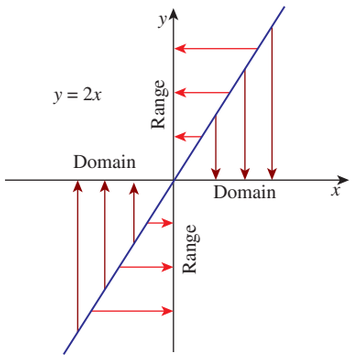


Figure 1.9

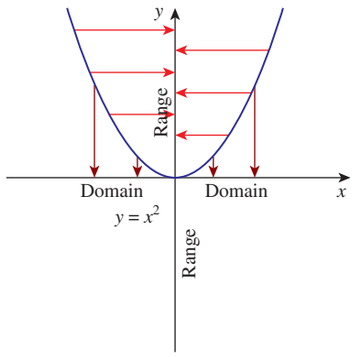


Figure 1.10

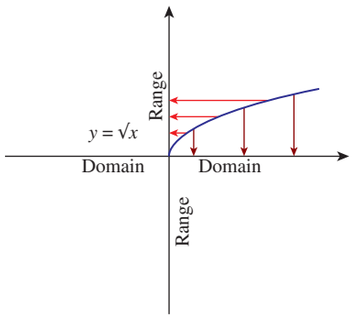


Figure 1.11

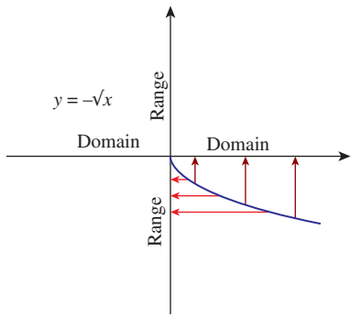


Figure 1.12

Note that, the domain of a relation is a subset of the first set in the Cartesian product and the range is a subset of second set. Usually we call the second set as ***co-domain*** of the relation. Thus, the range of a relation is the collection of all elements in the co-domain which are related to some element in the domain. Let us note that the range of a relation is a subset of the co-domain.

For any set A, $\emptyset$ and $A\times A$ are subsets of $A\times A$. These two relations are called ***extreme relations***. The former relation is an ***empty relation*** and the later is an ***universal relation***.

We will discuss more about domain, co-domain and the range in the next section namely, "Functions".

If R is a relation from A to B and if $(x,y)\in R$, then sometimes we write xRy (read this as "x is related to y") and if $(x,y)\notin R$, then sometimes we write $x\not{R}y$ (read this as "x is not related to y").

Though the general definition of a relation is defined from one set to another set, relations defined **on** a set are of more interest in mathematical point of view. So let us concentrate on relations defined on a set.